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Proceedings of the Combustion Institute 33 (2011) 929–937
Proceedings of the Combustion Institute
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Effect of methane-dimethyl ether fuel blends on flame stability, laminar flame speed, and Markstein length

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Available online 19 August 2010

Abstract

Binary fuel blends provide challenges to chemical kinetics models not seen by testing only pure fuels, allowing them to be proven over a wider range of inputs. The effect of adding dimethyl ether (DME) to methane (CH_4) on the laminar flame speed, Markstein length, and Lewis number was studied experimentally and numerically over a range of initial pressures from 1 to 10 atm, with the volumetric ratios of the fuel blends ranging from 60% CH_4/40% DME to 80% CH_4/20% DME. This data set includes high-pressure results that have never been published before. The experimental results were compared to an improved kinetics model, an ongoing effort spanning the past few years. Model results are in very good agreement with the experimental data. For the 80/20 blend of CH_4/DME, the Lewis number remained close to unity as the equivalence ratio increased, in comparison to the large decrease in Lewis number for pure DME as equivalence ratio is increased. This small change in Lewis number, with the value remaining near unity, resulted in the 80/20 blend of CH_4/DME remaining stable throughout the entire range of 5-atm experiments, while all other pure fuels and blends exhibited instabilities at initial pressures equal to or greater than 5 atm. In addition, the Markstein lengths were greatly affected by the blending of the fuels. A small amount of DME addition caused the Markstein lengths to change by a large value. Finally, a rigorous uncertainty analysis was performed on the experimental data, giving the error with respect to the true value rather than the standard deviation of repeated experiments.
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Keywords: Laminar flame speed; Methane; DME; Lewis number; High-pressure

1. Introduction

The laminar flame speed and Markstein length have been extensively studied in the past for pure methane-air flames [1–8], and studies on pure dimethyl ether (DME) are also widely available [9–15]. However, studies on CH_4/DME binary fuel blends are scarce [16]. Also, there are few chemical kinetics models that accurately predict laminar flame speed in addition to high-temperature shock-tube and low-temperature rapid compression machine (RCM) ignition properties for

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doi:10.1016/j.proci.2010.05.042

pure and binary blended fuels. The authors' kinetic model has been verified for C1–C3 alkanes at high pressure for pure fuels and their blends for laminar flame speed and high-temperature shock-tube and low-temperature RCM ignition target data [1,17–19], for the laminar flame speed of pure DME at elevated pressures [15], and for other hydrocarbons [20,22].

The purpose of this study was twofold: (1) validate a revised chemical kinetics model for binary mixtures of an alkane and an ether and (2) provide new experimental data at elevated pressure for binary mixtures of CH_4 and DME. This study presents laminar flame speed data for 1-, 5-, and 10-atm initial pressures for binary fuel blends of 60% CH_4/40% DME and 80% CH_4/20% DME, as well as their Markstein lengths and Lewis numbers. The new experimental data are compared to the pure fuels that compose them, as well as to the authors' chemical kinetics model. Finally, the effect of blending the fuels on the Lewis number and the resulting flame stability are also discussed.

2. Experimental approach

2.1. Facility and data analysis

A full, detailed description of the experimental facility is available in Lowry et al. [1] and de Vries [23], so only a brief overview is provided here. The constant-pressure, outwardly propagating spherical flame was used in this study to collect the experimental data. The large internal size of the vessel, 30.5 cm internal diameter and 35.6 cm internal length, allowed flame radii up to 4.6 cm to be used in the data analysis without any significant deviation from constant-pressure behavior [24]. The constant-pressure assumption is further validated by Fig. 1, where it is shown that there is less than a 1% increase in pressure during the time when measurements are taken. In this study, the radii used for analysis ranged from 0.8 cm up to 4.6 cm, effectively avoiding both ignition disturbances and deviation from constant pressure.

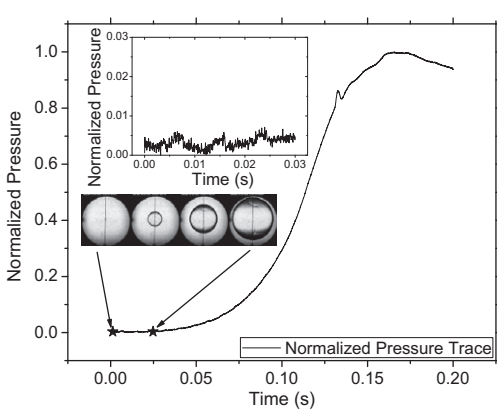

Fig. 1. Experimental pressure trace showing time period of measurement. The inset figure shows a closer view of the constant-pressure measurement time period.

Mixtures were made using the partial pressure method via 0–1000 Torr and 0–500 psi pressure transducers, and vacuum pressures were measured using a Varian 0–2000 mTorr transducer. The gas purities used in this study were Grade 2.6 (99.6%) for DME, Grade 3.7 (99.97%) for CH_4, and UHP (99.999%) for both O_2 and N_2. The mixtures were centrally ignited by creating a spark across two electrodes. The current and voltage across the electrodes are continuously adjustable to minimize the ignition energy for each experiment. The data were taken using a Z-type schlieren system with a high-speed digital camera capturing the resulting images at 1500 frames/s. Figure 2 shows example images obtained in this study.

The laminar flame speed and Markstein length were found by analyzing the images recorded by the high-speed camera. The radius at each time step was found by using a best-fit algorithm to fit a circle to each recorded image. The unstretched, burned flame speed and burned Markstein length were found by performing linear regression on the radii and time steps, using the integrated Markstein relationship given by Eq. (1) [25–27].

$$R_f = S^0_{L,b}t - 2L_{m,b}\ln(R_f) + B \tag{1}$$

In the previous relation, R_f is the flame radius, $S^o_{L,b}$ is the burned, un-stretched flame speed, $L_{m,b}$ is the burned Markstein length, B is an integration constant, and t is the time. The unburned, unstretched flame speed, $S^o_{L,u}$, and the unburned Markstein length, $L_{m,u}$, are found by dividing the burned values by the ratio of the unburned gas density (reactants) to the burned gas density (products). Residuals, the difference between experimental radii and the predicted radii, are plotted versus flame radius to determine where the flame becomes unstable. The predicted radii are solved for using an iterative method. First, the flame speed, Markstein length, and offset are found using linear regression on the experimental data. Then, the experimental flame speed, Markstein length, and offset are input to Eq. (1), the integrated Markstein relation. In Eq. (1), all the values are now known except for the time and radius at each time step. Finally, the predicted radius is found by iteratively adjusting the radius until the resulting time matches each experimental time step to within 0.000001 s. The predicted radius is now compared to the experimentally measured radius, shown by Fig. 3. Since the residuals are initially centered about zero, the flame can be assumed to propagate linearly. Additionally, flame acceleration is easily seen using this method. When the residuals are no longer

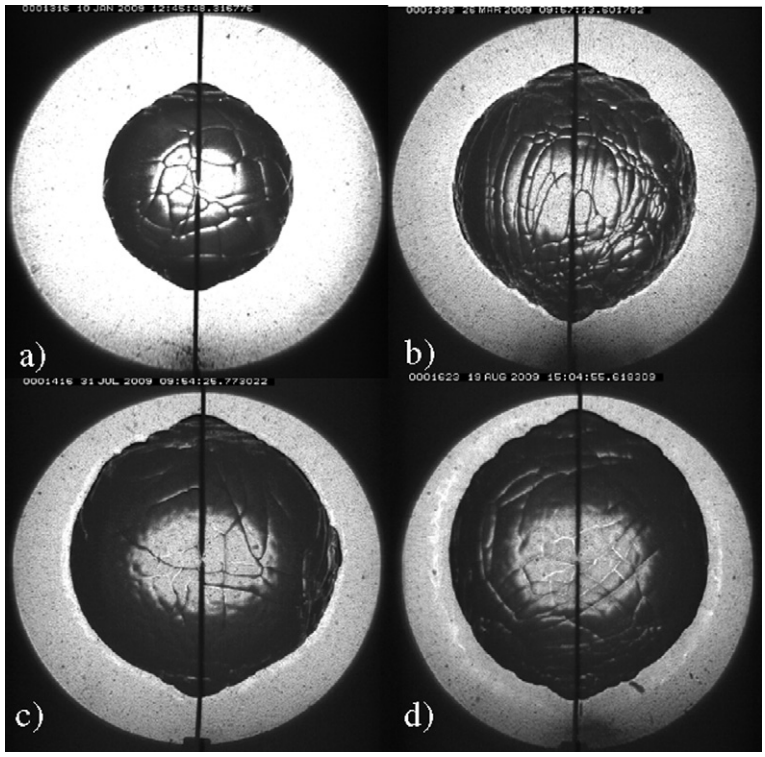

Fig. 2. Examples of flames at their critical radii for stoichiometric 5-atm experiments: (a) methane [1], (b) DME [15], (c) 60/40 CH_4/DME, and (d) 80/20 CH_4/DME (did not become unstable). The values of critical radius are given in Table 1.

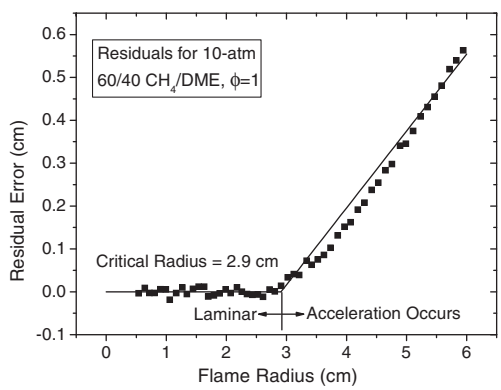

Fig. 3. An example of a residuals plot showing flame acceleration.

centered about zero, the flame is considered unstable. This method provides a quantitative way to determine where the flame becomes unstable. In contrast, determining flame acceleration via the recorded images is a subjective process that may give different results depending on the person performing the analysis.

A thorough uncertainty analysis was performed for the data presented herein, as outlined by Moffat [28]. The results of the uncertainty analysis are presented in Section 3. The bias (systematic) error is given by Eq. (2), where B_{S_L} is the total systematic error, u_i is the fixed uncertainty for each variable x_i, and $\frac{\partial S^o_{L,u}(x_i)}{\partial x_i}$ is the partial derivative of the flame speed, $S^o_{L,u}$, with respect to the variable x_i.

$$B_{S_L} = \left(\sum_{i=1}^{n} \left(\frac{\partial S^o_{L,u}(x_i)}{\partial x_i} u_i \right)^2 \right)^{1/2} \tag{2}$$

From the above definition, a relationship between the flame speed and each variable must be known. A functional relationship of the form $S^o_{L,u} = (a + b * \phi + c * \phi^2) * (1/P_i)^{(d+e*\phi+f*\phi^2)}$ is proposed, where the coefficients are determined from experimental data and are presented as Supplementary data to the present paper. This relational form fits the experimental data extremely well. The random uncertainties in this study are described by Eq. (3).

$$S_{yx} = \left(\frac{\sum_{i=1}^{N} (y_i - y_{ci})^2}{v} \right)^{1/2} \tag{3}$$

Here y_i is the experimental value, y_{ci} is the predicted value from the mth order polynomial, N is the total number of data points, and v is the

degrees of freedom, $N-(m+1)$. The precision interval at a 95% confidence level becomes $P_{S_L} = \pm t_{v,0.95} S_{yx}$.

The total uncertainty, found by combining B_{S_L} and P_{S_L} using the root sum square method, ranged from 2.5% to 10.5% with the higher percentage uncertainty occurring at higher pressure. A characteristic uncertainty is shown in Fig. 4.

2.2. Lewis number calculation

Lewis number, a dimensionless ratio of the thermal to mass diffusivity, can give valuable insight into the flame response for a given mixture [29–31]. The calculations presented herein are not intended to be the most accurate determination of the Lewis number (*Le*), but are provided to show the general trends in *Le* resulting from blending CH_4 and DME. The method used in this study to calculate *Le* is the ratio of the thermal diffusivity of the mixture over the mass diffusivity of the deficient species into the diluent, as shown in Eq. (4) where λ is the mixture thermal conductivity, c_p is the mixture specific heat, ρ is the mixture density, and $D_{i,j}$ is the mass diffusivity of the deficient species (i) into the diluent (j). The Lennard–Jones potential data for the binary fuel blends were approximated as the mass average of the pure-fuels' parameters.

$$Le = \frac{\lambda}{c_p \rho D_{i,j}}. \tag{4}$$

2.3. Kinetics model

The detailed chemical kinetics mechanism is based on the hierarchical nature of hydrocarbon combustion mechanisms containing the H_2/O_2 sub-mechanism [22], together with the CO/CH_4 and the C2 and C3 sub-mechanisms that have already been published [17–19]. The C4 sub-mechanism has been fully detailed in two recent papers on the butane isomers [20,21]. The DME sub-mechanism is based on Fischer et al. [32,33] with minor updates on the low-temperature chemistry that have no effect on flame speed predictions; this part of the work is ongoing and is not detailed in the present paper. One of the most sensitive reactions as far as laminar flame speed is concerned is $H + CH_3\ (+M) = CH_4\ (M)$ for both CH_4/air and DME/air systems and their binary mixtures; this reaction is pressure dependant and therefore becomes more sensitive at higher pressures. To get better agreement at high pressures, the rate constant of this reaction was revised in the present paper; specifically, the rate constant expression in the GRI 2.11 mechanism [34] has been taken as the basis, and both high- and low-pressure limit A factors were decreased within uncertainty limits. A sensitivity analysis for a 60/40 CH_4/DME mixture is presented and discussed in the Discussion section. The Premix [35] module in Chemkin Pro [36] was used to perform the kinetic model calculations. The version of the model used in the current work is C4_49, which is available online together with associated thermochemical parameters at <http://c3.nuigalway.ie/mechanisms.html>.

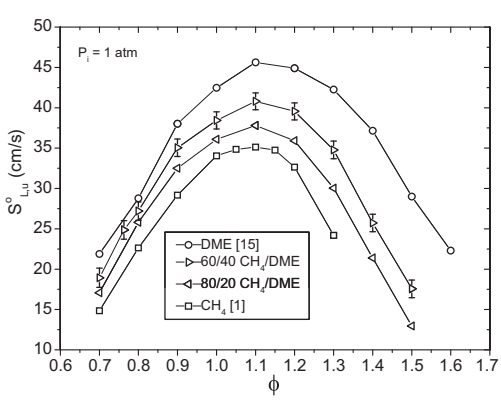

Fig. 4. Laminar flame speed for 1-atm pure DME [15], 60/40 CH_4/DME, 80/20 CH_4/DME, and pure CH_4 [1].

3. Results

Presented in this section are the laminar flame speed results for the binary blends of CH_4/DME at initial pressures of 1, 5, and 10 atm covering equivalence ratios from 0.7 to 1.6. The fuel blends consist of 60% CH_4/40% DME and 80% CH_4/20% DME by volume.

3.1. Mixture flame speeds compared to reference pure fuels

Flame speed as a function of equivalence ratio for the binary fuel blends is compared to the pure fuels that compose the blends in Figs. 4–6. Note that the lines between experimental points in Figs. 4–6 are to guide the eye and do not represent any additional data or the results of the kinetics model. The pure-fuel results (also taken from the same facility) are shown for comparison but are fully treated elsewhere [1,15]. Figure 4 presents the experimental flame speed results for DME, 60% CH_4/40% DME, 80% CH_4/20% DME, and pure CH_4, all at atmospheric pressure. Error bars on the 60% CH_4/40% DME data provide an uncertainty similar to that for all other results. The error bars are omitted for all other data for figure clarity.

It is seen that the flame speed of the mixture increases with increasing concentration of DME, almost linearly. The flame speed results for an initial pressure of 5 atm for the pure and blended fuels are shown in Fig. 5.

As expected, increasing the initial pressure reduces the flame speed. This trend is the case

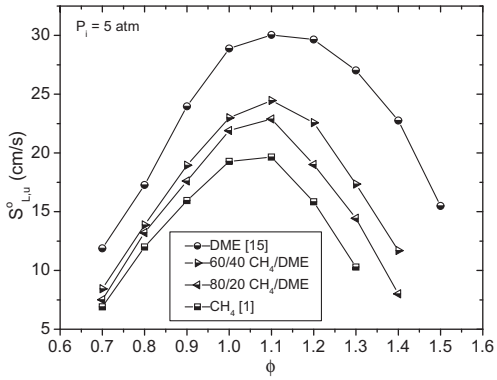

Fig. 5. Laminar flame speed for 5-atm pure DME [15], 60/40 CH_4/DME, 80/20 CH_4/DME, and pure CH_4 [1].

for all initial pressures herein. Figure 6 gives the results for all the fuels studied at an initial pressure of 10 atm. The results for pure DME were limited at an initial pressure of 10 atm due to instabilities [15].

3.2. Mixture flame speeds with model

In Figs. 7 and 8, the experimental data are compared to the chemical kinetics model described above. Figure 7 shows the model performance against the 60% CH_4/40% DME binary fuel blend for a variety of equivalence ratios and at initial pressures of 1, 5, and 10 atm.

Figure 7 shows that the model works well for binary fuel blends. The model appears to capture the chemistry, handles pressure increase, and is in near-perfect agreement with the experimental data at all pressures. In addition, it again shows that increasing the DME concentration increases the laminar flame speed. Figure 8 compares the data of Chen et al. [16] and the model to the experimental results for the 80% CH_4/20% DME binary fuel blend for initial pressures of 1, 5, and 10 atm.

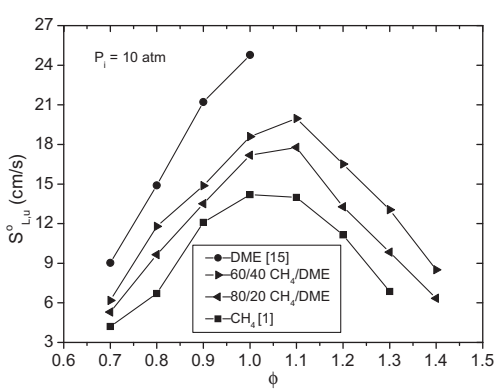

Fig. 6. Laminar flame speed for 10-atm pure DME [15], 60/40 CH_4/DME, 80/20 CH_4/DME, and pure CH_4 [1].

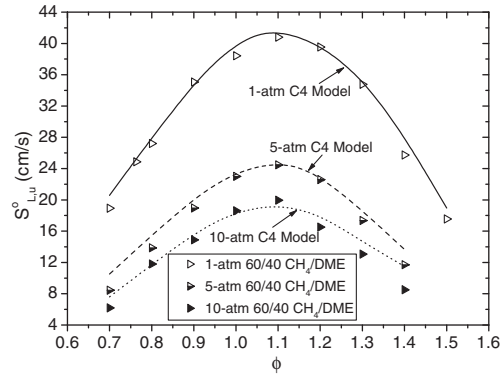

Fig. 7. Laminar flame speed for 60/40 CH_4/DME at 1, 5, and 10-atm initial pressure compared to the kinetics model.

Again, the model captures the chemical effects of blending the fuels, as shown in Figs. 7 and 8, and is in good agreement with the experimental data at all pressures. The new experimental data compare favorably to the limited data published by Chen et al. [16]. Finally, the overall effect of pressure for each fuel tested is shown in Fig. 9. The curves in Fig. 9 are empirical correlations of the pressure dependence for each mixture of the form $a(1/P)^n$, where a and n are constants, and P is the pressure in atm.

As shown by Fig. 9, the laminar flame speed of DME is the least sensitive to pressure, and the sensitivity of the laminar flame speed to pressure increases with increasing methane concentration. The relationships for the laminar flame speed as a function of pressure for each fuel are also given in Fig. 9.

4. Discussion

Instabilities were seen for all fuels in this study at an initial pressure of 10 atm. At 5-atm initial

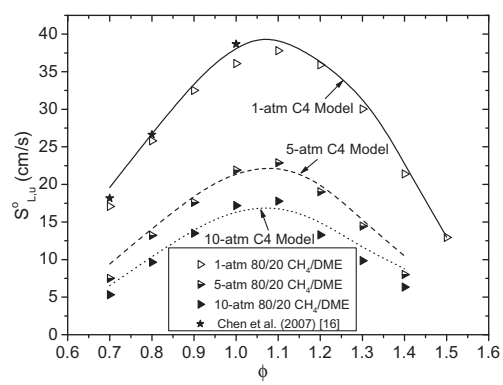

Fig. 8. Laminar flame speed for 80/20 CH_4/DME at 1, 5, and 10-atm initial pressure compared to other data and kinetics model.

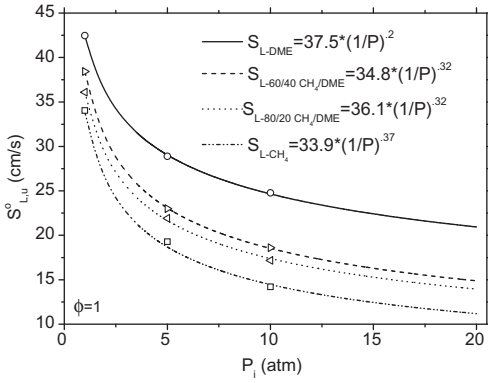

Fig. 9. A comparison of the laminar flame speed for the pure fuels and fuel blends studied herein with changing pressure.

pressure, instabilities were seen for pure CH_4, pure DME, and the 60% CH_4/40% DME at varying radii. Table 1 gives the critical radius, determined as outlined in the experimental approach section, for the fuels studied.

Table 1 shows that the radius where the flame becomes unstable depends on the fuel composition. Opposite trends are seen for the response of CH_4 and DME and this result is expected due to the opposite trends in Lewis number. Table 2 gives the Lewis numbers for the fuels studied.

As can be seen from Table 2, CH_4 and DME display opposite trends in Lewis number, and their binary blends display a combination of the pure-fuels' Lewis number trends. Overall, DME shows a greater change in the value of *Le* when compared to methane. The blends mirror this behavior as well; the 60/40 blend of CH_4/DME displays a larger change in Lewis number than the 80/20 blend. In addition, the stability of the flame tends to roughly follow the *Le* trends. For example, for pure DME, the radius where the flame becomes unstable occurs at increasingly smaller values as the Lewis number falls below unity. Smaller values of critical radius denote that the flame becomes unstable sooner in its development.

Table 1
Critical radii (cm) for CH_4, DME, and their binary blends at 5-atm initial pressure, where '–' denotes that the flame did not become unstable.

φ	CH_4	DME	60/40 CH_4/DME	80/20 CH_4/DME
0.7	3.98	–	–	–
0.8	4.29	4.89	–	–
0.9	4.19	4.75	4.83	–
1	3.30	4.23	4.65	–
1.1	3.57	3.72	4.16	–
1.2	4.40	3.17	4.75	–
1.3	4.52	2.77	–	–

Table 2
Lewis numbers for CH_4, DME, and their binary mixtures at STP.

φ	CH_4	DME	60/40 CH_4/DME	80/20 CH_4/DME
0.7	0.982	1.572	1.362	1.214
0.8	0.982	1.556	1.355	1.211
0.9	0.982	1.541	1.348	1.207
1.1	1.052	0.920	0.993	1.019
1.2	1.052	0.912	0.988	1.017
1.3	1.052	0.903	0.983	1.014

However, the critical radius values do not exactly match the *Le* trends. This behavior could be due to the small change in *Le* for methane when compared to DME, so its blends are not expected to show large changes in response to equivalence ratio changes. A fuel displaying a larger change in *Le* that followed the trend of CH_4, from above unity to below unity as equivalence ratio increases, would show more definitive trends in the critical radius in its pure form and in blends with other fuels displaying opposite trends in *Le*. The 5-atm data were used in the discussion of stability and the Lewis number because the 10-atm data did not clearly show trends relating the two. The authors feel that the transition from a laminar flame, marked by flame acceleration, occurs faster than the framing rate of the camera used in this study. Therefore, there is a sudden jump in the residuals plot, Fig. 3, when the flame becomes unstable for the 10-atm cases, and the radius exactly where this occurs is not able to be accurately defined. Defining this radius is possible for the 5-atm data because, even though the flame itself is faster, the transition to an unstable flame occurs more slowly. In addition, 10-atm pure DME became unstable very early in its propagation, and it was not possible to solve for the flame parameters for any of the fuel rich experiments due to the very small critical radii.

The Markstein length, a measure of the sensitivity of the effect of stretch on burning velocity, depends on the components of the mixture being investigated. Therefore, one would expect to see large differences in the Markstein length for the fuel blends compared to the pure fuels. The Markstein length is very sensitive to the number of images used in the analysis and depends heavily on the initial flame propagation. Due to the large size of the facility used herein, many images outside of the initial propagation disturbances can be captured for analysis. Care was taken to ensure that the first few flame images were spherical and exhibited no disturbances due to the electrodes, spark gap, or initial spark energy. The spark gap was set for each fuel blend mixture to minimize initial disturbances in the flame. Figure 10 shows Markstein lengths of pure CH_4, DME, 60/40 CH_4/DME, and 80/20 CH_4/DME as a function

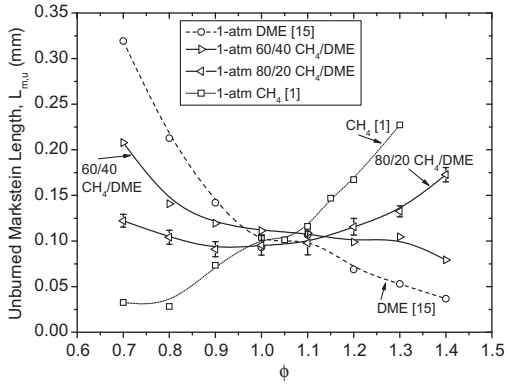

Fig. 10. Markstein lengths for DME [15], 60/40 CH_4/DME, 80/20 CH_4/DME, and CH_4 [1] at 1 atm.

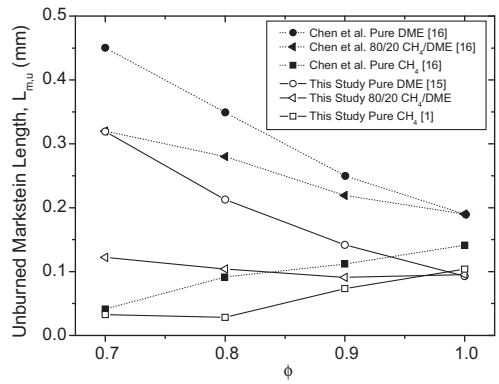

Fig. 11. Markstein lengths for DME [15], 80/20 CH_4/DME, and CH_4 [1] at 1-atm compared to the results of Chen et al. [16].

of equivalence ratio at an initial pressure of 1 atm. Also included in Fig. 10 is a characteristic Markstein length uncertainty, derived mainly from the regression statistics. The lines in Figs. 10 and 11 represent no additional data or model calculation, but are provided to clearly show trends and guide the eye.

Figure 10 shows that a small amount of DME added to CH_4 results in a large increase in the Markstein length. This was also observed by Chen et al. [16]. The dependence of the Markstein length on equivalence ratio for the fuels studied matches the expected trends derived from the *Le* trends. For the Markstein length, a positive dependence on equivalence ratio suggests that the *Le* depends positively on the equivalence ratio and vice versa [24,37]. This trend is seen for all the fuels studied, but it is not as clear for the 80/20 CH_4/DME blend due to the small change in *Le* and Markstein length as equivalence ratio changes. Figure 11 shows a comparison of the Markstein lengths found in this study with the results of Chen et al. [16].

From Fig. 11, the results for the Markstein lengths agree with Chen et al. [16] qualitatively, following the same trends. However, quantitatively, the results are different. The Markstein length results of Chen et al. [16] are consistently larger than the results from this study. The Markstein lengths for the pure-fuel cases have been compared to other published data [1,15] and they agree well, which set the upper and lower limits, adding confidence to the results presented herein. Additionally, the fuel rich Markstein length results for the CH_4/DME blends are the first ever published.

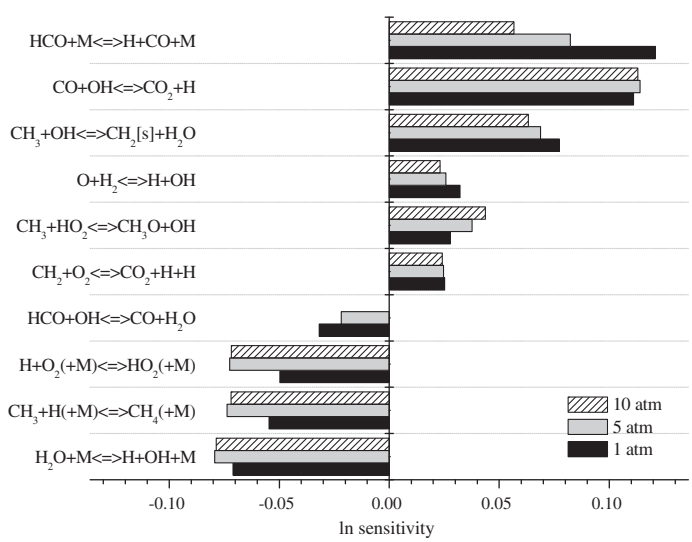

Fig. 12. Flame speed sensitivity analysis for a 60/40 CH_4/DME mixture with an equivalence ratio of $\varphi = 1.0$ at pressures of 1, 5 and 10 atm. The sensitivity of the reaction $H + O_2 = OH + O$ is the dominant one but is not shown to accentuate the sensitivities of the other reactions.

A sensitivity analysis was performed for a 60/40 CH_4/DME mixture at $\varphi = 1.0$, shown in Fig. 12. In general, the system is most sensitive to the high-temperature chain branching reaction $H + O_2 \rightleftarrows O + OH$; however, this reaction is omitted on the plot in Fig. 12 for sake of clarity. Reactions with positive sensitivity coefficients are promoting reactivity, while ones with negative coefficients are inhibiting. Recombination of CH_3 and H into methane is an inhibiting reaction competing with the chain branching for H atoms, which the system is very sensitive to at all pressures. Adjusting the rate constant of this reaction improved our results significantly especially at 5 and 10 atm conditions, as mentioned above.

5. Summary

This paper presents experimental results for binary blends of CH_4/DME, with volumetric fractions of 60% CH_4/40% DME and 80% CH_4/20% DME performed at initial pressures of 1, 5, and 10 atm. The elevated-pressure data are the first published for these blends of CH_4/DME. The experimental data are also compared to the authors' modified chemical kinetics model. Model agreement with the data is excellent in all cases. The model has been proven increasingly robust, agreeing with laminar flame speed data ranging from pure and blended C1–C3 alkanes to pure DME and its blends with CH_4. In addition, the model has been historically proven to agree extremely well with high-pressure shock-tube data at varying temperatures and RCM low-temperature data. A rigorous uncertainty analysis was performed, revealing an experimental uncertainty ranging from 2.5% to 10.5% of the true value of the laminar flame speed. Atmospheric Markstein lengths were also presented for all fuels studied, with discrepancies being seen between the current data and Chen et al. [16]. Finally, flame stability, determined by the critical radius, was experimentally shown to follow the *Le* trends of the mixture. At an initial pressure of 5 atm, all the fuels studied displayed decreasing critical radii as the *Le* decreased below unity and increasing critical radii as *Le* increased above unity. This behavior was especially evident in the fuel blends when comparing the 5-atm 60% CH_4/40% DME, where the *Le* decreased to below unity, to the 80% CH_4/20% DME, where the *Le* did not fall below unity.

Acknowledgments

This work was primarily supported by Rolls Royce Canada and additionally by SFI (Science Foundation Ireland) through grant number CHEF845. The authors gratefully acknowledge the assistance of Jaap de Vries in performing some of the flame speed experiments.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.042.

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